CEBU INSTITUTE OF TECHNOLOGY
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IT342 - G1
System Integration and Architecture

System Design Document (SDD)

Project Title: VetEase

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Version	Date	Author	Changes Made	Status
0.1	02/21/2026	Monteclaro, Kurt David M.	Initial draft	Review
1.0	2/28/2026	Monteclaro, Kurt David M.	Finalized SDD. Added OAuth, external API, file upload, email integration, updated API contract, database design, ERD, and UI wireframes.	Final

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EXECUTIVE SUMMARY

1.1 Project Overview

VetEase is a veterinary reservation and scheduling platform that enables pet owners to book clinic appointments online for services such as check-up, vaccination, grooming, deworming, consultation, and follow-up visits. The system provides a convenient booking experience for clients while helping veterinary clinics:

- Organize and visualize appointment schedules
- Reduce walk-in congestion and overlapping bookings
- Maintain structured appointment and pet records for future reference

1.2 Objectives

The primary objectives of VetEase are to:

- Provide online booking with real-time availability so pet owners can only select open time slots.
- Support role-based access control, distinguishing between Client (pet owner) and Clinic Admin/Staff roles.
- Implement a 3-tier architecture:
    - Backend: Spring Boot REST API (authentication, business logic, persistence)
    - Web Frontend: React SPA
    - Mobile Frontend: Kotlin Android app
- Offer full CRUD operations for:
    - Pet profiles
    - Clinic services
    - Clinic schedules (hours, slot length, blocked dates)
    - Appointments (create, view, update, cancel, reschedule)
- Deliver a complete, working prototype that is realistic and maintainable for solo development within one academic semester, while remaining extensible for future enhancements.

1.3 Scope

Included Features:

- **Authentication & Authorization**
    - User registration, login, and logout using JWT-based authentication.
    - Role-based access (Client vs Admin/Staff).

- **Client Dashboard**
    - Display of the next upcoming appointment.
    - Overview of appointment history.
- **Pet Profile Management**
    - Add, edit, and delete pet profiles.
    - Optional fields for vaccine history and notes (e.g., medical conditions, behavior).
- **Clinic Services Management**
    - Client-facing list of clinic-defined services (consultation, check-up, vaccination, grooming, deworming, follow-up, etc.).
    - Admin management of service catalog (add/update/deactivate) if time permits.
- **Appointment Booking**
    - Selection of pet, service, date, and time slot.
    - Availability checker that generates time slots based on clinic hours and slot length.
    - Prevention of double-booking for the same service/date/time combination.
- **Appointment Management**
    - Client:
        - View upcoming and past appointments
        - Cancel appointments
        - Request reschedule.
    - Status handling:
        - Pending
        - Confirmed
        - Completed
        - Cancelled

- **Admin Panel**
    - View and confirm/approve pending appointment requests.
    - View today's schedule and filter by date/status.
    - Manage clinic hours and slot configuration.
    - Define and manage blocked dates (holidays, special closures).

Excluded Features:

- Online payment integration (cards, e-wallets, etc.).
- SMS notifications excluded. Basic email notifications included (welcome + appointment confirmation).
- Telemedicine / video consultation capabilities.
- Full electronic medical records (EMR) including detailed medical history, lab results, and prescriptions.
- Multi-clinic or marketplace support (initial version is designed for a single clinic instance only).

1.0 INTRODUCTION

1.1 Purpose

This System Design Document (SDD) defines the overall design of VetEase, a veterinary appointment reservation and scheduling system. It describes:

- Functional and non-functional requirements
- System architecture and technology stack
- API contracts and integration patterns
- Database schema and data relationships
- UI/UX design direction (web and mobile)
- Implementation and delivery plan

2.0 FUNCTIONAL REQUIREMENTS SPECIFICATION

2.1 Project Overview

- **Project Name:** VetEase
- **Domain:** Veterinary Clinic Scheduling / Appointment Booking
- **Primary Users:**
    - **Pet Owners (Clients) –** register, manage pets, book appointments
    - **Clinic Staff/Admin –** manage schedule, confirm appointments, configure clinic settings
- **Problem Statement:**
    - Veterinary clinics experience walk-in congestion, manual logging of appointments, and double-booked time slots.
    - Pet owners lack a simple way to book verified appointment slots and review their appointment history.
- **Solution:**
- A slot-based appointment booking system where:
    - Clients can see only available time slots and book them.
    - Admin can confirm, complete, or cancel appointments and configure clinic hours and blocked dates.
    - Both web and mobile clients access the same backend API.

2.2 Core User Journeys

Journey 1: First-time Client Booking

1. Client opens a web or mobile app.
2. Registers an account (optionally as admin if meant for clinic staff).

3. Logs in using username and password.
4. Adds a pet profile (name, species/breed, age, notes, optional vaccine history).
5. Goes to the Book Appointment page.
6. Chooses:
    - A pet from their pets
    - A service (e.g., Check-up, Vaccination, Grooming, Deworming, Consultation)
    - A date
7. The system calls the availability API and shows only open time slots.
8. Client selects an available slot and adds notes (symptoms/concerns).
9. Client submits the booking.
10. Appointment is created with status = PENDING.
11. Admin views pending requests and confirms it -> status becomes CONFIRMED; it appears on today's schedule when the date matches.

Journey 2: Returning Client Booking / Management

1. Client logs in.
2. Go to My Appointments page.
3. Sees upcoming appointments and past history.
4. For an upcoming appointment:
    - **Cancel**: status → CANCELLED, slot becomes available again.
    - **Reschedule**: selects new date & slot; system checks availability and updates appointment (status reset to PENDING).
5. Client can also:
    - Update pet profiles
    - Book new appointments

Journey 3: Administrator Schedule Management

- Admin logs in (account with role = ADMIN).
- Open Admin Panel.

- Views:
    - Pending appointments list (all future PENDING).
    - Today's schedule (appointments for today).
- For each appointment:
    - Confirm → status = CONFIRMED
    - Complete → status = COMPLETED
    - Cancel → status = CANCELLED
- In Settings:
    - Configures clinic opening/closing time and slot length (e.g., 30 minutes).
- In Blocked Dates:
    - Adds/removes blocked dates (holidays/no service days) so no bookings can be made on those days.

2.3 Feature List (MoSCoW)

MUST HAVE

- **Authentication**
    - Register, login, logout
    - Role support: CLIENT and ADMIN
- **Pet Profile Management (CRUD)**
    - Add/edit/delete pets for logged-in client
- **Clinic Services List**
    - Client view of active services
    - Default services seeded: Check-up, Vaccination, Grooming, Deworming, Consultation
- **Appointment Booking with Slot Availability**
    - Only show open slots for a date/service
    - Prevent double booking of the same service/time slot

- **Appointment Management**
  - Client: view own appointments, cancel, request reschedule
  - Status: PENDING, CONFIRMED, COMPLETED, CANCELLED
- **Admin Panel**
  - Admin confirmation / completion / cancellation of appointments
  - View today's schedule & pending list
  - Basic clinic settings & blocked dates
- **External API Integration**
  - The system retrieves pet breed data from a public external API.
  - The breed list is displayed in the pet profile form.
- **Google OAuth Authentication**
  - Users can log in using their Google account.
  - The system creates or links user accounts and issues JWT.
- **File Upload for Pet Profile**
  - Users can upload pet images.
  - Images are stored on the server and linked to pet records.
- **Email Notification System**
  - Send welcome email upon account registration.
  - Send an appointment confirmation email when the admin approves the appointment.

SHOULD HAVE

- Configurable clinic hours and slot duration
- Blocked dates (holidays/days-off)
- Clear validation & user-friendly error messages

COULD HAVE

- More structured vaccine history per pet
- Search/filter appointments by date and status

- Admin notes on appointments

WON'T HAVE

- Online payments
- Multi-clinic or multi-branch support

2.4 Detailed Feature Specifications

Feature: User Authentication

- **Screens (Web/Mobile):**
  - Register
  - Login
- **Fields (current project):**
  - username (used for login)
  - email
  - password
  - firstName
  - lastName
  - role (CLIENT or ADMIN) – selectable on registration
- **Backend Endpoints:**
  - POST /api/auth/register
    - Body: { username, email, password, firstName, lastName, role? }
    - Creates a user with role CLIENT by default; ADMIN allowed when explicitly chosen.
  - POST /api/auth/login
    - Body: { username, password }
    - Returns { token, username, email, firstName, lastName, role }
  - GET /api/user/me
    - Authenticated; returns UserDto (id, username, email, firstName, lastName, role)
- **Security**:
  - Passwords hashed with BCrypt
  - JWT tokens issued on login; attached via Authorization: Bearer <Token>
  - Protected routes: all /api/** except POST /api/auth/register, POST /api/auth/login, and GET /api/services

Feature: Pet Profile Management

- **Client Capabilities:**
  - Add/edit/delete pets tied to logged-in user.

- **Fields:**
    - name (required)
    - species (e.g., Dog, Cat)
    - breed
    - age (optional integer)
    - notes (optional)
    - vaccineHistory (optional, free-text)
- **Backend Endpoints:**
    - GET /api/pets
        - Returns list of the current user's pets (PetDto).
    - POST /api/pets
        - Creates a new pet belonging to the authenticated user.
    - PUT /api/pets/{id}
        - Updates pet fields; enforces that the pet belongs to the current user.
    - DELETE /api/pets/{id}
        - Deletes the pet; enforced ownership.

Feature: Clinic Services List

- **Client View (Web/Mobile):**
    - List of services with name, description, and duration (minutes).
- **Backend Endpoints:**
    - GET /api/services (public)
        - Returns list of active services.
- (Optional future admin endpoints – currently not implemented but design-ready):
    - POST /api/admin/services
    - PUT /api/admin/services/{id}

- DELETE /api/admin/services/{id}

Feature: Appointment Booking (Transactional)

- **Inputs (Client):**
  - petId
  - serviceId
  - date (YYYY-MM-DD)
  - time (HH:mm / HH:mm:ss)
  - notes (symptoms/concerns)
- **System Rules:**
  - Only allow booking on dates not blocked and within clinic hours.
  - For a given date + time + service, reject new bookings if there is an existing appointment with status in {PENDING, CONFIRMED}.
- **Backend Endpoints:**
  - GET /api/availability?date=YYYY-MM-DD&serviceId={id}
    - Returns list of available time strings (["09:00:00", ...]) generated from clinic settings minus taken slots.
  - POST /api/appointments
    - Body (CreateAppointmentRequest): { petId, serviceId, date, time, notes }
    - Checks:
      - Pet belongs to current user
      - Service exists & active
      - Date not blocked
      - Slot not already occupied
      - Creates Appointment with status PENDING.

Feature: Appointment Management

- **Client Side:**
  - GET /api/appointments/mine
    - Lists all appointments for logged-in user, sorted by date/time.
  - POST /api/appointments/{id}/cancel
    - Cancels own appointment (if not already completed/cancelled).
  - POST /api/appointments/{id}/reschedule?date=YYYY-MM-DD&time=HH:mm:ss
    - Reschedules own appointment; re-validates availability and resets status to PENDING.

- **Admin Side:**
    - GET /api/admin/appointments/today
        - Today's schedule.
    - GET /api/admin/appointments/pending
        - All pending future appointments.
    - POST /api/admin/appointments/{id}/confirm
    - POST /api/admin/appointments/{id}/complete
    - POST /api/admin/appointments/{id}/cancel
- **Status Lifecycle:**
    - PENDING → (admin confirm) → CONFIRMED
    - PENDING/CONFIRMED → (admin/client cancel) → CANCELLED
    - CONFIRMED → (admin complete) → COMPLETED
    - Reschedule resets to PENDING.

Feature: Admin Panel

- **Settings:**
    - Clinic hours: openingTime, closingTime
    - Slot length: slotMinutes
    - Stored in ClinicSettings (single row).
- **Blocked Dates:**
    - GET /api/admin/blocked-dates
    - POST /api/admin/blocked-dates?date=YYYY-MM-DD
    - DELETE /api/admin/blocked-dates/{id}
- **Role Check:**
    - Admin endpoints all verify User.role == "ADMIN" server-side.

Feature: External API Integration – Pet Breed Lookup

The system integrates with a public external API to retrieve pet breed data dynamically. This allows users to select valid breeds when creating or editing pet profiles.

- **User Capabilities**

    - Client can:
        - View breed dropdown when adding/editing pet
        - Select breed retrieved from external API

Backend Endpoint : GET /api/external/dog-breeds

Returns list of breeds retrieved from Dog API.

System Flow

1 Frontend requests breed list
2 Backend calls external API
3 Backend returns breed list
4 Frontend displays dropdown

Validation

API failures handled gracefully
Fallback allows manual entry

Feature: Google OAuth Authentication

Users can log in using their Google account. The system verifies the Google token and generates its own JWT token.

- **User Capabilities**
    - User can:
        - Click Login with Google
        - Access system without manual registration

Backend Endpoint : POST /api/auth/oauth/google

System Rules

- If user does not exist:
    - Create new user:
        - If exists:
            - Login existing user
            - Generate JWT token
- **Security**
    - Google token verified server-side
    - JWT used for system authorization

Feature: Pet Image Upload

Users can upload images for their pets. Images are stored on the server and linked to the pet record.

- **User Capabilities**
  - Client can:
    - Upload pet image
    - View uploaded image
  - Admin can:
    - View pet image

Backend Endpoint : POST /api/pets/{id}/upload-photo

System Rules

- User must own pet
- Only image files allowed
- File size limited

Storage
Images stored in: /uploads/pets/

Feature: Email Notification System

The system sends automated email notifications for important events.

Email Types

- Welcome Email
- Sent after user registration.
- Appointment Confirmation Email
- Sent when the admin confirms the appointment.

System Components

- **EmailService**
  - Uses SMTP server.

Trigger Events

- User registration success

- Admin confirms appointment

2.5 Acceptance Criteria

AC-1: Successful Registration

- Given I am a new user
- When I register with valid username, email, password, and names
- Then my account is created with role CLIENT (or ADMIN if chosen), and I can log in and access the dashboard.

AC-2: No Double Booking

- Given a specific service/date/time slot already has an appointment with status PENDING or CONFIRMED
- When another client attempts to book that same service/date/time
- Then the system rejects the booking with an appropriate error and keeps the original appointment only.

AC-3: Admin Confirmation

- Given an appointment is in PENDING status
- When an admin confirms it from the Admin Panel
- Then its status becomes CONFIRMED and it appears in today's schedule when the appointment date is today.

AC-4: Google Login

- Given a user logs in using Google
- When authentication succeeds
- Then system logs in user and issues JWT

AC-5: Pet Image Upload

- Given user uploads pet image
- When upload is valid
- Then image is stored and linked to pet

AC-6: Email Notification

- Given appointment is confirmed
- When admin confirms appointment
- Then confirmation email is sent

3.0 NON-FUNCTIONAL REQUIREMENTS

3.1 Performance

- API response time: most requests (95%) should complete within ≤ 2 seconds under demo load.
- Support at least 50–100 concurrent users (sufficient for lab-scale demo).
- Use JPA/Hibernate with indexes on:
    - appointments.date, appointments.time
    - Foreign keys: appointments.client_id, appointments.pet_id, appointments.service_id

3.2 Security

- JWT-based stateless authentication.
- Role-based authorization:
    - Normal users cannot access /api/admin/**.
- Passwords hashed with BCrypt.
- Use JPA/Hibernate (no manual SQL strings) to protect against SQL injection.
- Recommended for deployment (not necessarily in lab):
    - HTTPS termination (e.g., reverse proxy / hosting platform).
    - Basic rate limiting at API gateway or reverse proxy level.

3.3 Compatibility

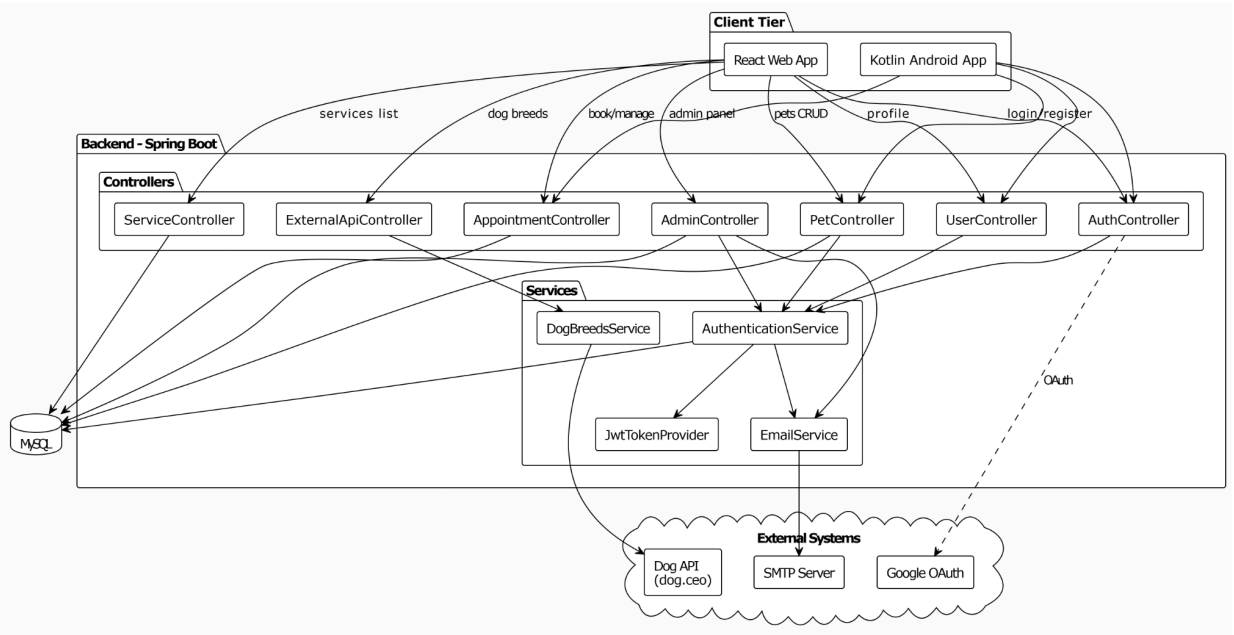
- **Backend:**
    - Java 21, Spring Boot (3.x), MySQL 8.0
- **Web app:**
    - Chrome/Edge/Firefox (latest)
    - Responsive layout from mobile up to desktop
- **Mobile:**
    - Android API 24+ (Android 7.0+)
    - Emulator support (using 10.0.2.2) and physical devices (LAN IP)

3.4 Usability

- Booking flow should be complete within ~2 minutes by a first-time user.
- Clear validation messages:
    - Required fields
    - Slot already taken
    - Date blocked / clinic closed
- Dark, modern VetEase UI with:
    - Clear navigation (dashboard, pets, services, appointments, admin)
    - Distinct status badges for PENDING, CONFIRMED, COMPLETED, CANCELLED.

4.0 SYSTEM ARCHITECTURE

4.1 Component Diagram



Technology Stack:

- **Web Frontend (React)**
    - Routes: /login, /register, /dashboard, /pets, /services, /appointments, /appointments/book, /admin
    - **Components:**
        - AuthContext
            - Stores authenticated user information
            - Stores JWT access token
            - Provides login, logout, and Google OAuth login methods
        - ProtectedRoute
            - Restricts access to authenticated users
            - Restricts access based on user role (CLIENT / ADMIN)
        - AppShell
            - Shared layout wrapper
            - Sidebar navigation
            - Header with user info and logout
        - Feature pages:
            - LoginPage
                - Username/password login
                - Google OAuth login button
            - RegisterPage
                - User account registration
            - DashboardPage
                - Displays next appointment
                - Displays profile summary
            - PetsPage
                - Add/edit/delete pets
                - Upload pet image
                - Breed dropdown populated from external API
            - ServicesPage
                - Displays clinic services
            - BookAppointmentPage
                - Select pet, service, date, time slot
                - Shows availability from backend
            - AppointmentsPage
                - Displays user appointment list
                - Cancel and reschedule appointments
            - AdminPanelPage
                - View pending appointments

            - Confirm, complete, cancel appointments
            - Configure clinic settings
            - Manage blocked dates
        - **Utilities**
            - api.js
                - Axios instance configured with base URL
                - Automatically attaches JWT token
            - oauth.js
                - Handles Google OAuth login flow
            - fileUpload.js
                - Handles multipart image upload
            - Axios REST API requests
                - Multipart file upload requests
            - External Integrations
                - Google Identity Services (OAuth login

- **Mobile App (Android, Kotlin)**
    - **Platform**:
        - Android API Level 34
    - **Architecture**:
        - MVVM (Model-View-ViewModel)
    - **Activities**:
        - LoginActivity
            - Username/password login
            - Google OAuth login
        - RegisterActivity
            - User registration
        - DashboardActivity
            - Displays next appointment
        - PetsActivity
            - Displays pet list
            - Upload pet images
        - BookAppointmentActivity
            - Booking flow with slot selection
        - AppointmentsActivity
            - Displays appointment list
            - Cancel and reschedule
        - AdminActivity (ADMIN role only)
            - Confirm and manage appointments

    - **Networking**:

        - RetrofitClient
            - Configured with backend base URL
            - Sends JWT token in Authorization header
    - **API Services:**
        - AuthApiService
        - PetApiService
        - AppointmentApiService
        - AdminApiService
        - ExternalApiService
    - **Image Upload:**
        - Uses MultipartBody.Part
        - Uploads images via Retrofit
    - **Token Storage:**
        - SharedPreferences
        - Stores JWT securely
    - **External integrations:**
        - Google OAuth via Google Sign-In SDK
- **Backend (Spring Boot REST API)**
    - **Framework:**
        - Spring Boot
    - **Architecture Pattern:**
        - Layered Architecture
    - **Layers:**
        - Controller Layer
        - Service Layer
        - Repository Layer
        - Security Layer
    - **Controllers**
        - AuthController
            - Register user
            - Login user
            - Google OAuth login
            - Logout
        - UserController
            - Retrieve authenticated user (/me endpoint)
        - PetController
            - Create pet
            - Update pet
            - Delete pet
            - Upload pet image

        - ServiceController
            - Retrieve clinic services
        - AppointmentController
            - Create appointment
            - Cancel appointment
            - Reschedule appointment
        - AdminController
            - Confirm appointment
            - Complete appointment
            - Cancel appointment
            - Manage clinic settings
        - BlockedDateController
            - Add/remove blocked dates
        - ExternalApiController
            - Retrieve breed data from external API
- **Services**
    - AuthenticationService
        - Register user
        - Authenticate user
        - Generate JWT
    - OAuthService
        - Verify Google OAuth token
        - Create or retrieve OAuth user
    - JwtTokenProvider
        - Generate JWT tokens
        - Validate tokens
    - PetService
        - Manage pet records
    - AppointmentService
        - Manage appointments
        - Validate availability
    - AvailabilityService
        - Generate available slots
    - ClinicSettingsService
        - Manage clinic hours
    - BlockedDateService
        - Manage blocked dates
    - ExternalApiService
        - Fetch pet breed data from public API

    - FileStorageService
      - Store uploaded pet images
      - Retrieve file paths
    - EmailService
      - Send welcome email
      - Send appointment confirmation email
- **Security**
    - SecurityConfig
      - Configures authentication rules
      - Defines protected routes
    - JwtAuthenticationFilter
      - Validates JWT token
    - PasswordEncoder
      - BCrypt password hashing
    - OAuthTokenVerifier
      - Verifies Google ID tokens
    - Role-based authorization:
        - CLIENT role access restrictions
        - ADMIN role access restrictions
- **Persistence (JPA / Hibernate)**
    - **Entities:**
        - User
        - Pet
        - ClinicService
        - Appointment
        - ClinicSettings
        - BlockedDate
        - PetImage
        - OAuthAccount
    - **Repositories:**
        - UserRepository
        - PetRepository
        - ClinicServiceRepository
        - AppointmentRepository
        - ClinicSettingsRepository
        - BlockedDateRepository
        - PetImageRepository
        - OAuthAccountRepository
    - **DTOs:**
        - UserDt

        - PetDto
        - AppointmentDto
        - ServiceDto
    - **Data Seeder:**
        - DataSeeder
        - Inserts default services
        - Inserts default clinic settings
- **File Storage**
    - Uploads directory:
        - /uploads/pets/
    - Stores:
        - Pet images
        - Database stores file path reference.
- **Email Integration**
    - Uses:
        - Spring Boot JavaMailSender
    - SMTP provider:
        - Gmail SMTP or Mailtrap
    - Triggers:
        - User registration
        - Appointment confirmation
- **External API Integration**
    - Public API:
        - Dog Breed API
    - Accessed via:
        - RestTemplate or WebClient
    - Used for:
        - Breed selection in pet profiles
- **Database (MySQL)**
    - Relational Database:
    - MySQL 8.0
    - Stores:
        - users
        - pets
        - services
        - appointments
        - clinic_settings
        - blocked_dates
        - pet_images

    - Relationships:
        - User → Pets (One-to-Many)
        - User → Appointments (One-to-Many)
        - Pet → Appointments (One-to-Many)
        - Service → Appointments (One-to-Many)
        - Pet → PetImages (One-to-Many)
    - Indexes:
        - appointments(date, time, service_id)
        - pets(owner_id)
    - Managed using:
        - Spring Data JPA
    - Database tools:
        - MySQL Workbench
- **External Services**
    - Google OAuth 2.0
      – User authentication
    - Dog Breed Public API
      – Breed lookup
    - SMTP Email Server
      – Email notifications
- **Development Tools**
    - Backend:
        - IntelliJ IDEA
    - Frontend:
        - VS Code
    - Mobile:
        - Android Studio
    - Version Control:
        - Git
        - GitHub
    - Testing Tools:
        - Postman

5.0 API CONTRACT & COMMUNICATION

5.1 API Standards

Base URL	http://localhost:8080/api
Format	JSON request/response
Authentication	Authorization: Bearer <accessToken>

Response Structure	{
	"success": true,
	"data": { /* payload */ },
	"error": {
	"code": null,
	"message": null,
	"details": null
	},
	"timestamp": "2026-02-27T10:30:00Z"
	}

5.2 Endpoint Specifications

Authentication Endpoints

User Registration

Description	User Registration
API URL	/api/auth/register
HTTP Request Method	POST
Format	JSON for all requests/responses
Authentication	None

Request Payload	{
	"username": "drsmith",
	"email": "drsmith@clinic.com",
	"password": "P@ssw0rd123",
	"firstName": "John",
	"lastName": "Smith",
	"role": "ADMIN"
	}

Response Structure	{
	"success": true,
	"data": {
	"user": {
	"id": 12,
	"username": "drsmith",
	"email": "drsmith@clinic.com",
	"firstName": "John",

	"lastName": "Smith",
	"role": "ADMIN"
	}
	},
	"error": {
	"code": null,
	"message": null,
	"details": null
	},
	"timestamp": "2026-02-27T10:30:00Z"
	}

User Login

Description	User Login
API URL	/api/auth/login
HTTP Method	POST
Format	JSON for all requests/responses
Authentication	None

Request Payload	{
	"username": "petowner01",
	"password": "password123"
	}

Response Structure	{
	"success": true,
	"data": {
	"user": {
	"id": 25,
	"username": "petowner01",
	"email": "owner@example.com",
	"firstName": "Jane",
	"lastName": "Doe",
	"role": "CLIENT"
	},
	"accessToken": "<jwt-access-token>",
	"refreshToken":
	"<jwt-refresh-token-optional>"
	},
	"error": {
	"code": null,
	"message": null,
	"details": null
	},
	"timestamp": "2026-02-27T10:30:00Z"
	}

Pets

Description	List my pets, update pet, and delete pet
API URL	• /api/pets (for listing) • /api/pets/{id} (for updating and deleting)
HTTP Method	• GET • PUT • DELETE
Format	JSON for all requests/responses
Authentication	Bearer Token (CLIENT/ADMIN)
Response Data	PetDto[]
Description	Create Pet
API URL	/api/pets
HTTP Method	POST
Format	Format JSON for all requests/responses
Authentication	Bearer Token

	{
	"name": "Buddy",
	"species": "Dog",
	"breed": "Labrador",
	"age": 3,
	"notes": "Friendly and energetic",
	"vaccineHistory": "Rabies (2025-01-10), Distemper (2025-01-10)"
	}

Availability

Description	Check for a date available
API URL	/api/availability?date=YYYY-MM-DD&serviceId={id}
HTTP Method	GET
Format	JSON for all requests/responses
Authentication	None

Response Structure	{
	"success": true,
	"data": ["09:00", "09:30", "10:00", "10:30"],
	"error": { "code": null, "message": null, "details": null },
	"timestamp": "2026-02-27T10:30:00Z"
	}

Appointments (Client)

Description	Set an appointment, see earlier appointments, and cancel an appointment
API URL	/api/appointments
HTTP Method	• GET /api/appointments/mine (Bearer Token) • POST /api/appointments (Bearer Token) • POST /api/appointments/{id}/cancel (Bearer Token) • POST /api/appointments/{id}/reschedule?date=...&time=... (Bearer Token)
Format	JSON for all requests/responses
Authentication	Bearer Token

Appointment DTO Includes:	{
	"id": 123,
	"date": "2026-03-01",
	"time": "09:30",
	"status": "PENDING",
	"notes": "General check-up",
	"pet": {
	"id": 5,
	"name": "Buddy"
	},
	"service": {
	"id": 2,
	"name": "Check-up"
	},
	"client": null
	}

Google OAuth Login Endpoint

Description	Authenticate user using Google account
API URL	• /api/auth/oauth/google
HTTP Method	• POST
Format	JSON for all requests/responses
Authentication	None

Request Payload	{
	"idToken": "google-id-token"
	}

Response Structure	{
	"success": true,
	"data": {
	"user": {
	"id": 25,
	"username": "google_user",
	"email": "user@gmail.com",
	"firstName": "John",
	"lastName": "Doe",
	"role": "CLIENT"
	},
	"accessToken": "<jwt-access-token>"
	},
	"error": {
	"code": null,
	"message": null,
	"details": null
	},
	"timestamp": "2026-02-27T10:30:00Z"
	}

External API – Dog Breeds

Description	Retrieve dog breed list from external API
API URL	● /api/external/dog-breeds
HTTP Method	● GET
Format	JSON for all requests/responses
Authentication	None

Response Structure	{
	"success": true,
	"data": [
	"Labrador",
	"Golden Retriever",
	"Poodle"
	],
	"error": {
	"code": null,
	"message": null,
	"details": null
	},
	"timestamp": "2026-02-27T10:30:00Z"
	}

Upload Pet Image

Description	Upload image file for pet

API URL	• /api/pets/{id}/upload-photo
HTTP Method	• POST
Format	multipart/form-data
Authentication	Bearer Token
Request Payload	file: image file (jpg/png) Content-Type: multipart/form-data

Response Structure	{
	"success": true,
	"data": {
	"imageUrl": "/uploads/pets/pet123.jpg"
	},
	"error": {
	"code": null,
	"message": null,
	"details": null
	},
	"timestamp": "2026-02-27T10:30:00Z"
	}

Confirm Appointment (Admin)

Description	Admin confirms pending appointment
API URL	• /api/admin/appointments/{id}/confirm
HTTP Method	• POST
Format	JSON
Authentication	Bearer Token (ADMIN role required)
Request Payload	None

Response Structure	{
	"success": true,
	"data": {
	"status": "CONFIRMED"
	},
	"error": {
	"code": null,
	"message": null,
	"details": null
	},
	"timestamp": "2026-02-27T10:30:00Z"
	}

5.3 Error Handling

HTTP Status Codes

- 200 OK – Successful request.
- 201 Created – Resource created (e.g., new pet, new appointment).
- 400 Bad Request – Validation errors.
- 401 Unauthorized – Missing/invalid token.
- 403 Forbidden – Insufficient permissions (e.g., client hitting admin endpoint).
- 404 Not Found – Resource not found.
- 409 Conflict – Duplicate resource or double-booking attempt.
- 500 Internal Server Error – Unhandled exceptions.

Error Code Examples

Example 1	json
	{
	"success": false,
	"data": null,
	"error": {
	"code": "AUTH-001",
	"message": "Invalid credentials",
	"details": "Email or password is incorrect"
	},
	"timestamp": "2024-01-28T10:30:00Z"
	}

Example 2	{
	"success": false,
	"data": null,
	"error": {
	"code": "VALID-001",
	"message": "Validation failed",
	"details": {
	"email": "Email is required",
	"password": "Must be at least 8 characters"
	}
	},
	"timestamp": "2024-01-28T10:30:00Z"
	}

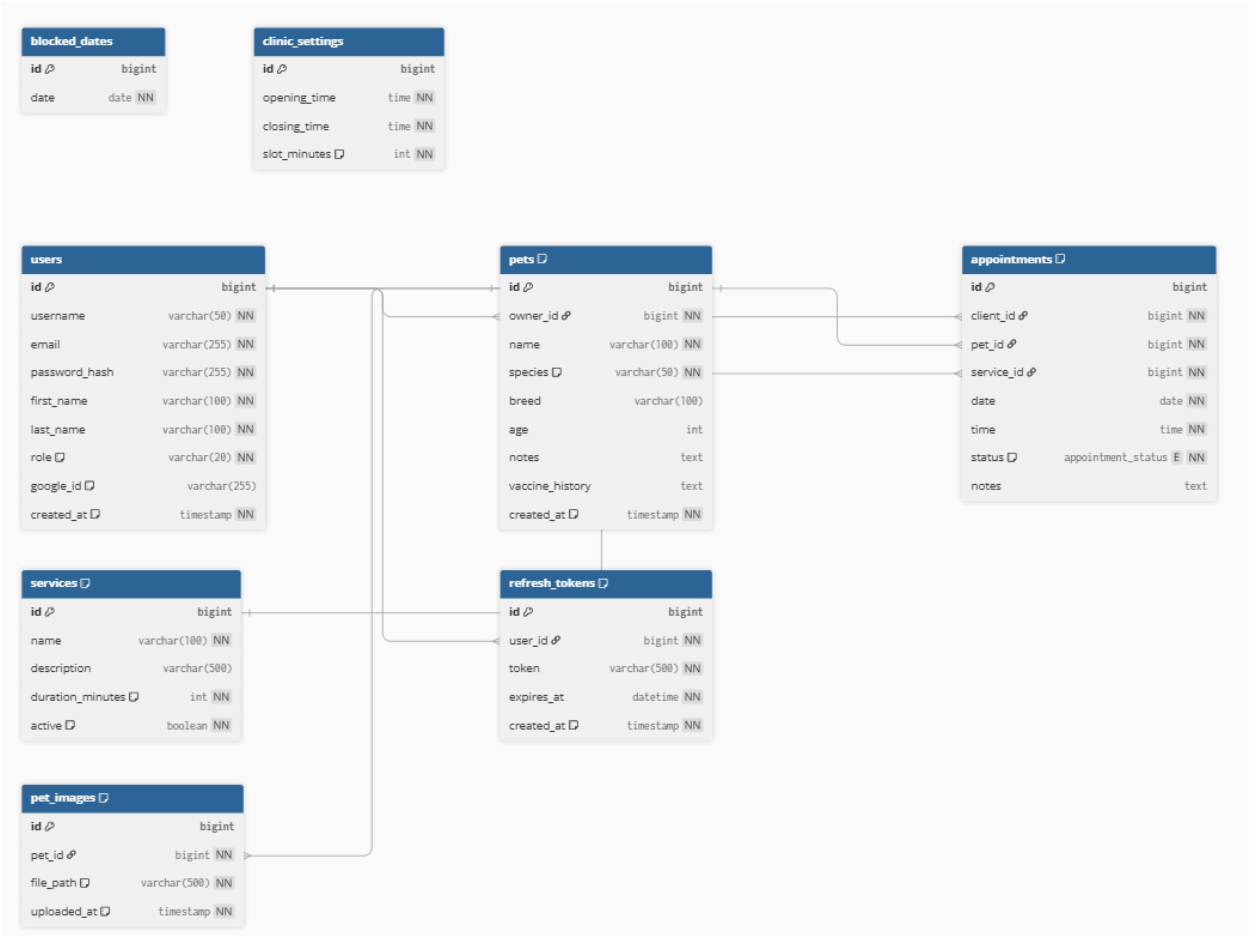
Common Error Codes

- AUTH-001: Invalid credentials

- AUTH-002: Token expired
- AUTH-003: Insufficient permissions
- VALID-001: Validation failed
- DB-001: Resource not found
- DB-002: Duplicate entry
- BUSINESS-001: Slot already booked
- SYSTEM-001: Internal server error

6.0 DATABASE DESIGN

6.1 Entity Relationship Diagram



Detailed Relationships:

- **User**

    - One User has many Pet records.
    - One User has many Appointment records (as client).
- **Pet**
    - Many Appointment records reference a single Pet.
- **ClinicService**
    - Many Appointment records reference a single ClinicService (service type).
- **Appointment**
    - Belongs to a User (client).
    - Belongs to a Pet.
    - Belongs to a ClinicService.
- **ClinicSettings**
    - Typically a single row containing global settings.
- **BlockedDate**
    - One row per blocked calendar date.

Key Tables:

- **Users**
    - id (PK)
    - username (unique)
    - email (unique)
    - password_hash
    - first_name
    - last_name
    - role (CLIENT / ADMIN)
    - created_at
- **Pets**
    - id (PK)
    - owner_id (FK → users.id)
    - name
    - species
    - breed
    - age
    - notes
    - vaccine_history
- **Services**
    - id (PK)
    - name (unique)
    - description
    - duration_minutes

    - active (boolean)

- **Appointments**
    - id (PK)
    - client_id (FK → users.id)
    - pet_id (FK → pets.id)
    - service_id (FK → services.id)
    - date (DATE)
    - time (TIME)
    - status (ENUM: PENDING, CONFIRMED, COMPLETED, CANCELLED)
    - notes (TEXT)
- **clinic_settings**
    - id (PK)
    - opening_time (TIME)
    - closing_time (TIME)
    - slot_minutes (INT)
- **blocked_dates**
    - id (PK)
    - date (DATE, unique)

Indexes are created on frequently queried columns:

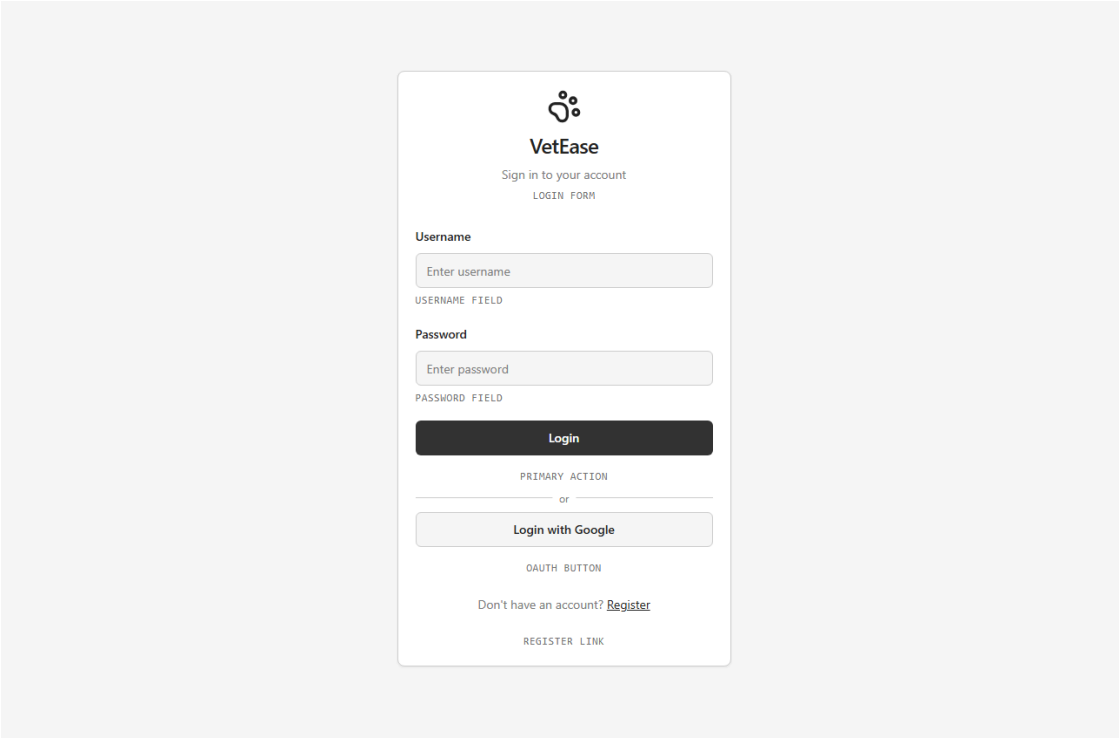
- appointments(date, service_id, time, status)
- pets(owner_id)
- services(active)

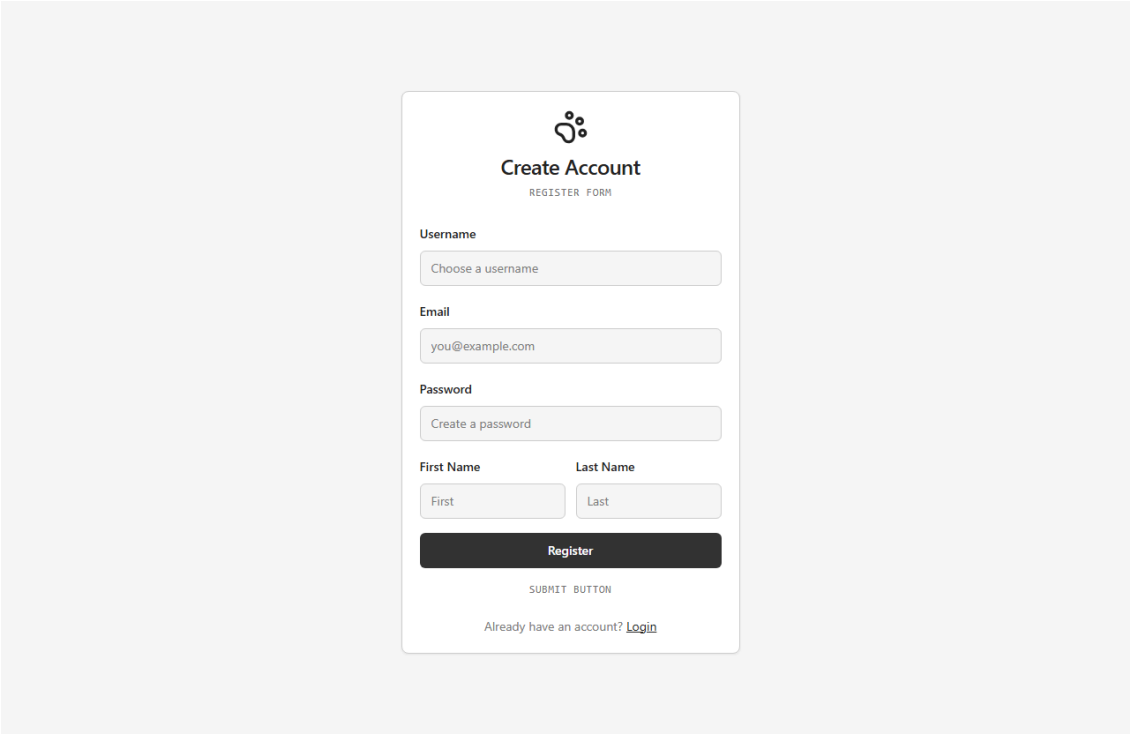
Table Structure Summary:

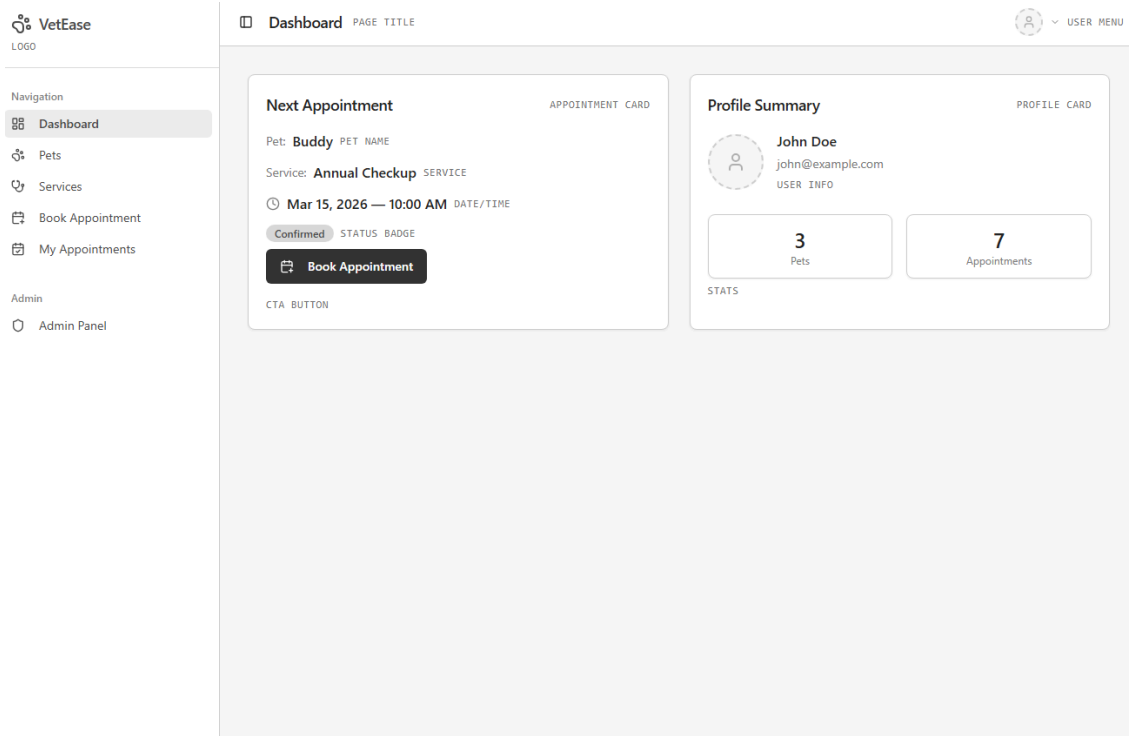
- users – accounts & roles
- pets – pet profiles per user
- services – clinic services offered
- appointments – bookings with status and times
- clinic_settings – opening/closing time and slot length
- blocked_dates – clinic closure days

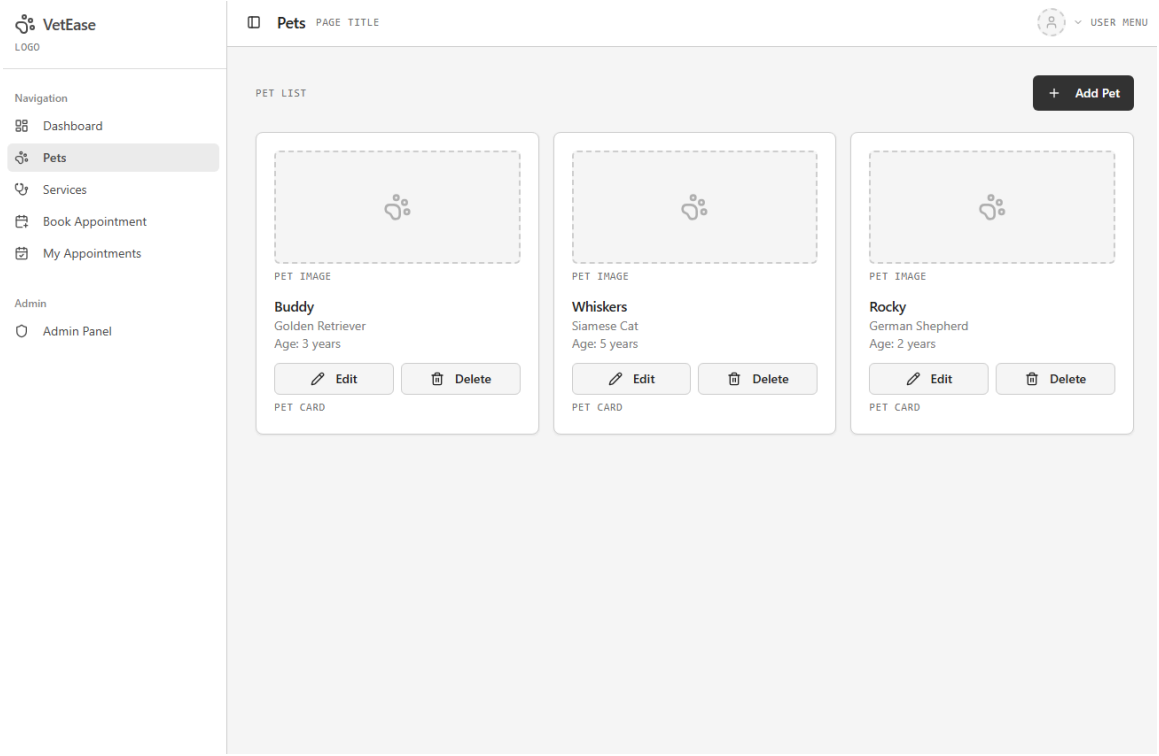
7.0 UI/UX DESIGN

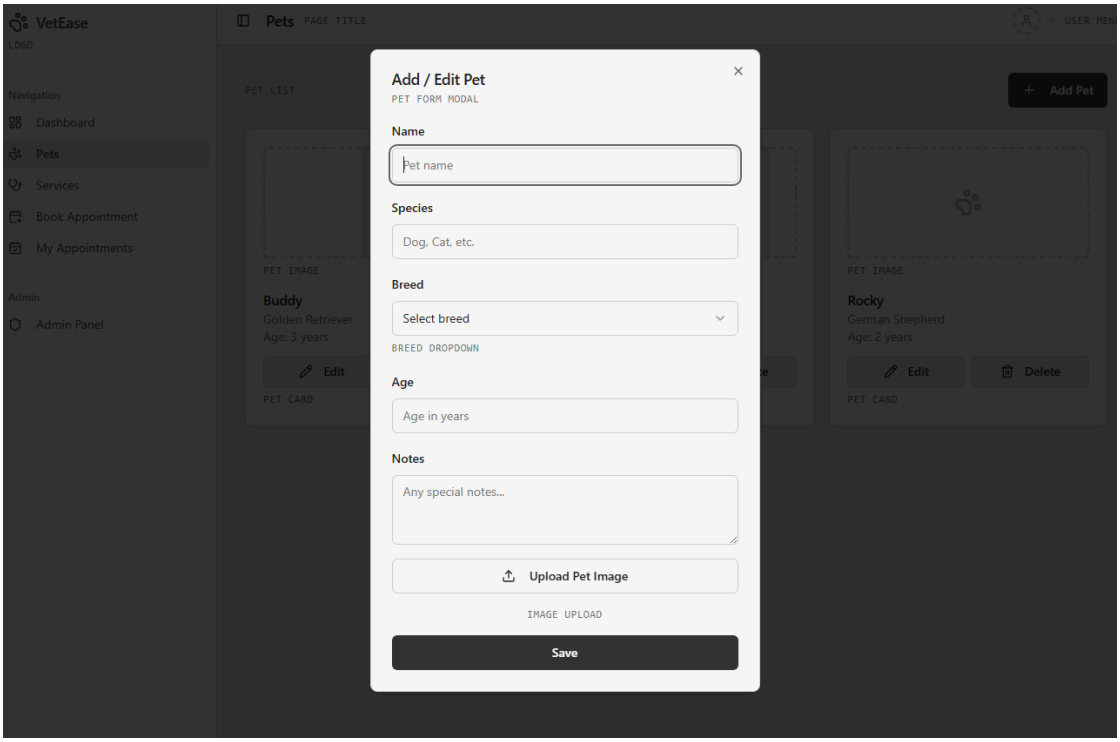
7.1 Web Application Wireframes

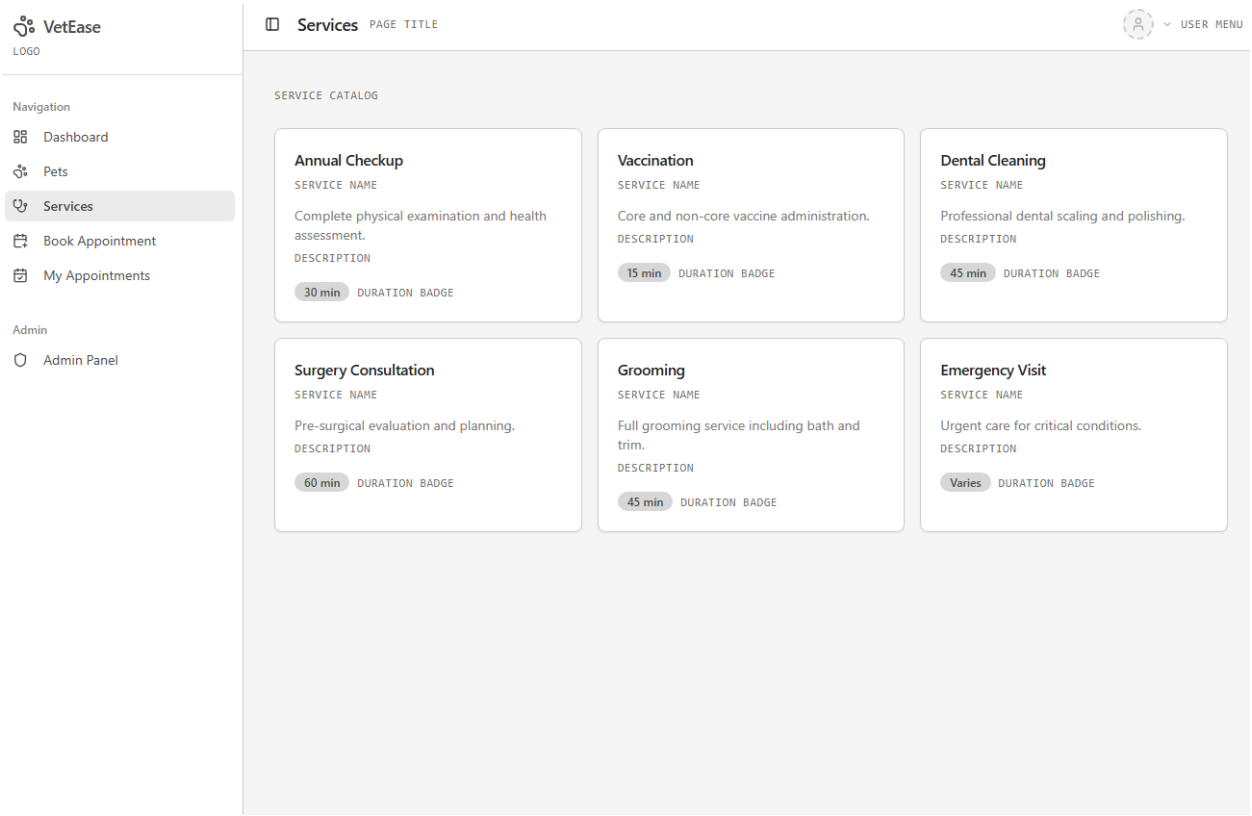





VetEase
LOGO
Navigation
Dashboard
Pets
Services
Book Appointment
My Appointments
Admin
Admin Panel
Dashboard
PAGE TITLE
USER MENU
Next Appointment
APPOINTMENT CARD
Pet: Buddy
PET NAME
Service: Annual Checkup
SERVICE
Mar 15, 2026 — 10:00 AM
DATE/TIME
Confirmed
STATUS BADGE
Book Appointment
CTA BUTTON
Profile Summary
PROFILE CARD
John Doe
john@example.com
USER INFO
3
Pets
7
Appointments
STATS


VetEase
LOGO
Navigation
Dashboard
Pets
Services
Book Appointment
My Appointments
Admin
Admin Panel
Pets
PAGE TITLE
USER MENU
PET LIST
Add Pet
PET IMAGE
Buddy
Golden Retriever
Age: 3 years
Edit
Delete
PET CARD
PET IMAGE
Whiskers
Siamese Cat
Age: 5 years
Edit
Delete
PET CARD
PET IMAGE
Rocky
German Shepherd
Age: 2 years
Edit
Delete
PET CARD


VetEase
LOGO
Pets
PAGE TITLE
USER MENU
Navigation
Dashboard
Pets
Services
Book Appointment
My Appointments
Admin
Admin Panel
PET LIST
Add Pet
PET IMAGE
Buddy
Golden Retriever
Age: 3 years
Edit
PET CARD
Add / Edit Pet
PET FORM MODAL
Name
Pet name
Species
Dog, Cat, etc.
Breed
Select breed
BREED DROPDOWN
Age
Age in years
Notes
Any special notes...
Upload Pet Image
IMAGE UPLOAD
Save
PET IMAGE
Rocky
German Shepherd
Age: 2 years
Edit
Delete
PET CARD


VetEase
LOGO
Services
PAGE TITLE
USER MENU
Navigation
Dashboard
Pets
Services
Book Appointment
My Appointments
Admin
Admin Panel
SERVICE CATALOG
Annual Checkup
SERVICE NAME
Complete physical examination and health assessment.
DESCRIPTION
30 min
DURATION BADGE
Vaccination
SERVICE NAME
Core and non-core vaccine administration.
DESCRIPTION
15 min
DURATION BADGE
Dental Cleaning
SERVICE NAME
Professional dental scaling and polishing.
DESCRIPTION
45 min
DURATION BADGE
Surgery Consultation
SERVICE NAME
Pre-surgical evaluation and planning.
DESCRIPTION
60 min
DURATION BADGE
Grooming
SERVICE NAME
Full grooming service including bath and trim.
DESCRIPTION
45 min
DURATION BADGE
Emergency Visit
SERVICE NAME
Urgent care for critical conditions.
DESCRIPTION
Varies
DURATION BADGE

VetEase
LOGO

Navigation
- Dashboard
- Pets
- Services
- Book Appointment
- My Appointments

Admin
- Admin Panel

Book Appointment
PAGE TITLE

USER MENU

BOOKING FORM

Select Pet

Choose a pet

PET DROPDOWN

Select Service

Choose a service

SERVICE DROPDOWN

Date

Pick a date

DATE PICKER

Notes

Any special requirements...

NOTES TEXTAREA

Book Appointment DISABLED UNTIL VALID

Available Time Slots

TIME SLOT GRID

9:00 AM	9:30 AM	10:00 AM
10:30 AM	11:00 AM	11:30 AM
1:00 PM	1:30 PM	2:00 PM
2:30 PM	3:00 PM	3:30 PM
4:00 PM	4:30 PM	

VetEase
LOGO

Navigation
- Dashboard
- Pets
- Services
- Book Appointment
- My Appointments

Admin
- Admin Panel

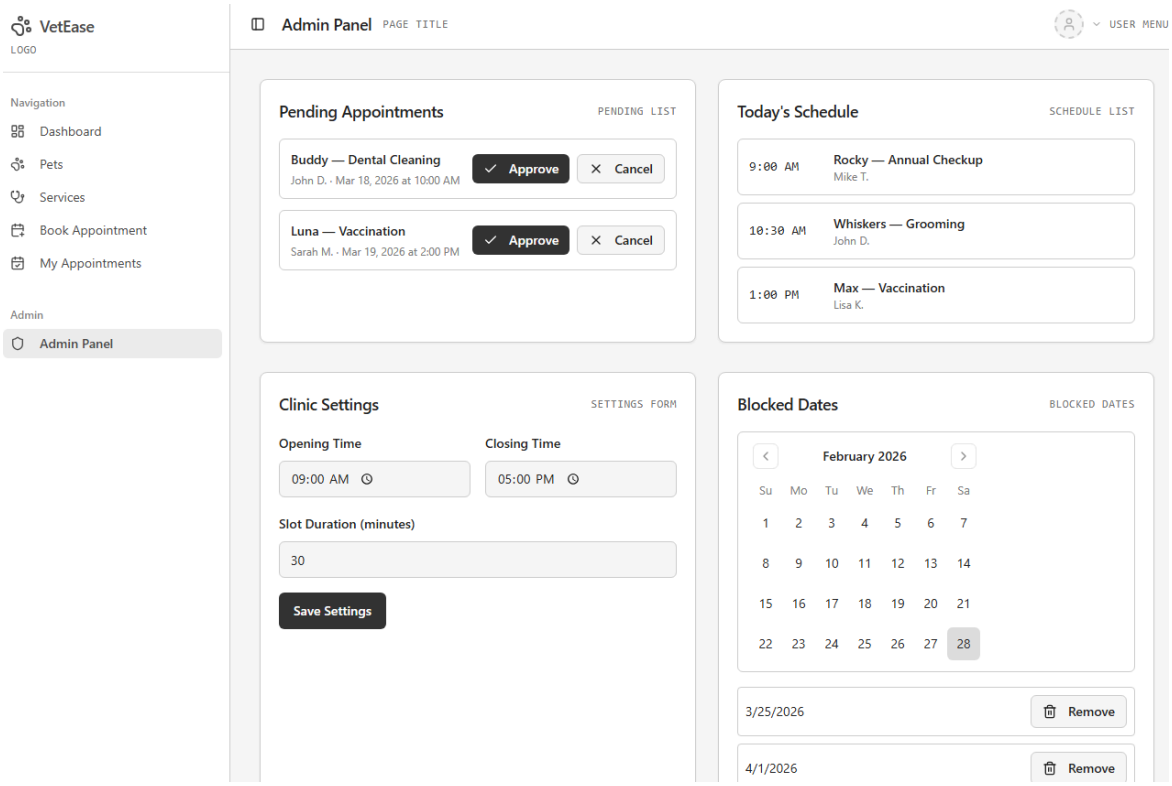
My Appointments
PAGE TITLE

USER MENU

APPOINTMENTS LIST

Date	Time	Pet	Service	Status	Actions
Mar 15, 2026	10:00 AM	Buddy	Annual Checkup	Confirmed	Reschedule Cancel
Mar 20, 2026	2:30 PM	Whiskers	Vaccination	Pending	Reschedule Cancel
Apr 02, 2026	11:00 AM	Rocky	Dental Cleaning	Confirmed	Reschedule Cancel
Jan 10, 2026	9:00 AM	Buddy	Grooming	Completed	Reschedule Cancel

RESPONSIVE TABLE → CARDS ON MOBILE



Global Layout

- Header or sidebar:
  - Brand (VetEase logo)
  - Navigation:
    - Dashboard
    - Pets
    - Services
    - Book Appointment
    - My Appointments
    - Admin (for admins only)
  - User profile mini-card + Logout button.

Dashboard (Client)

- Top Card: “Next Appointment”
  - Service name
  - Pet name
  - Date & time
  - Status badge (Pending/Confirmed)
  - CTA buttons: “Book New”, “View All”

- Profile Card:
    - Name, email, role (Client/Admin).
    - Quick links to Pets and Appointments.

Pets Page

- Left: Pet Form (Add / Edit)
    - Fields: Name, Species, Breed, Age, Notes, Vaccine History.
    - Buttons: Save, Cancel.
- Right: Pet List
    - Cards or rows:
        - Pet name
        - Species - Breed - Age
- Actions: Edit, Delete.

Services Page

- Grid of Service Cards:
    - Name
    - Duration chip (e.g., 30 min)
    - Description
- For admin extension: each card's options: Edit / Deactivate.

Book Appointment Page

- Step-like layout on a single page:
    - Step 1: Select Pet (dropdown)
    - Step 2: Select Service (dropdown)
    - Step 3: Select Date (date picker)
    - Step 4: Available Time Slots (button grid)
    - Step 5: Notes (textarea)
- CTA: "Submit Booking"

Disabled button until all selections are made.

My Appointments Page

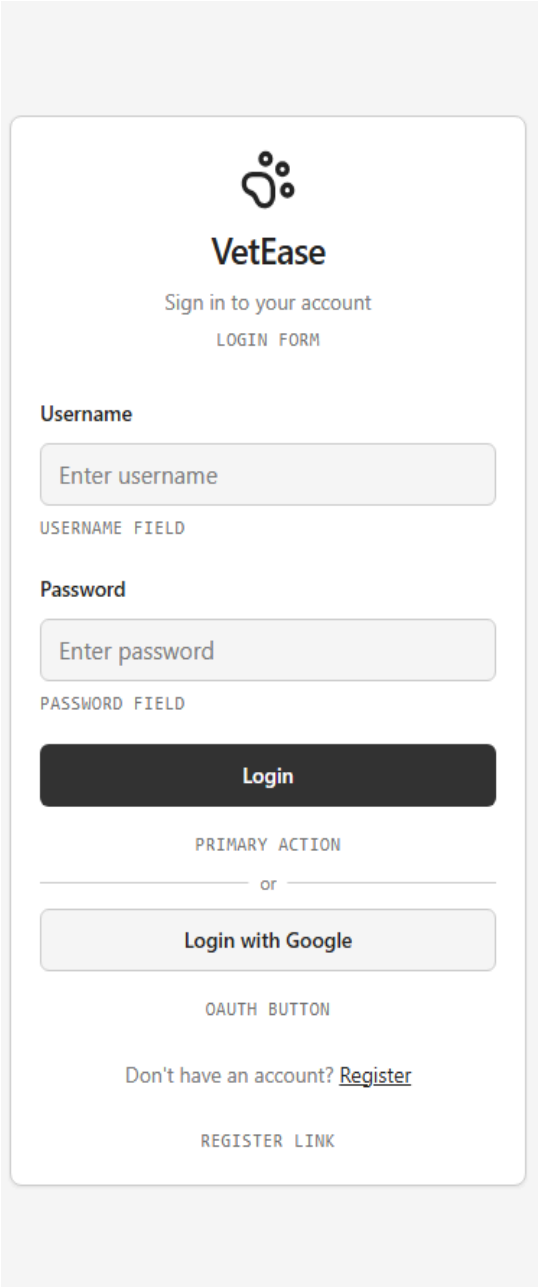
- List/table with filters:
    - Columns: Date, Time, Pet, Service, Status, Actions.
    - Actions:
        - **Reschedule** (opens mini panel with date + available slots).
        - **Cancel** (confirmation dialog).

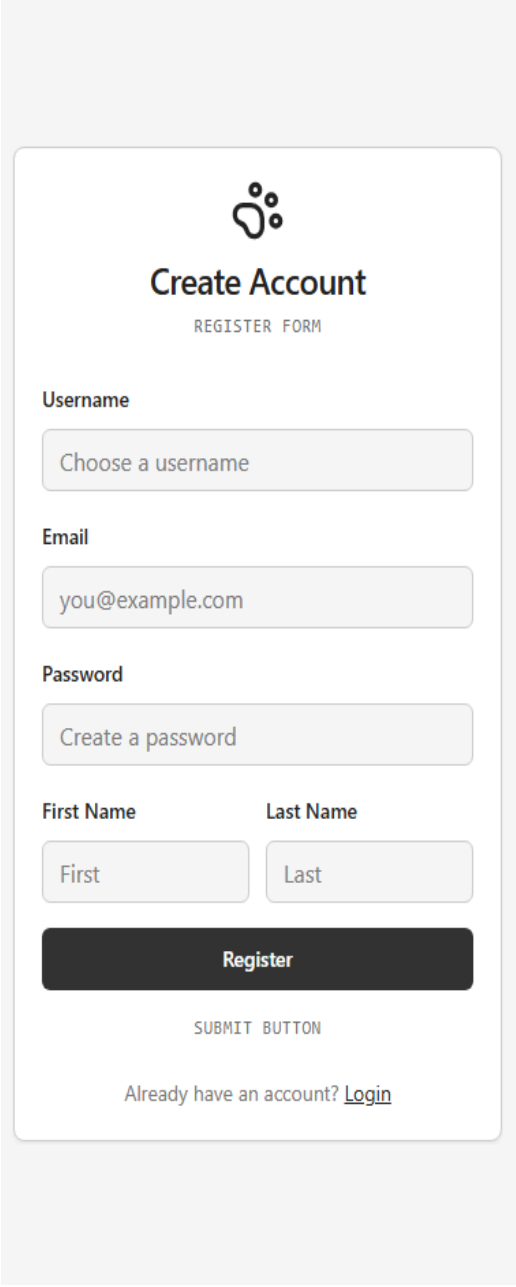
Admin Panel

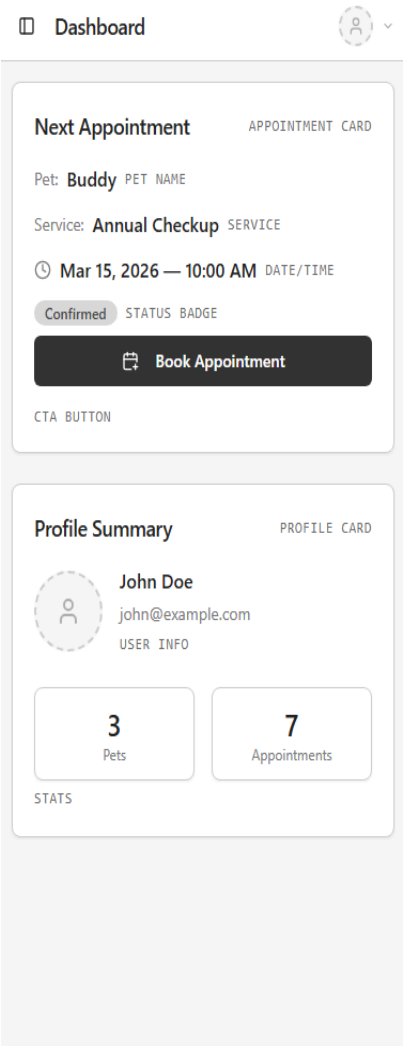
- Pending Requests:
  - Each row: Date, Time, Service, Pet, Owner, Status.
  - Actions: Confirm, Cancel.
- Today's Schedule:
  - Today's appointments; actions: Complete, Cancel.

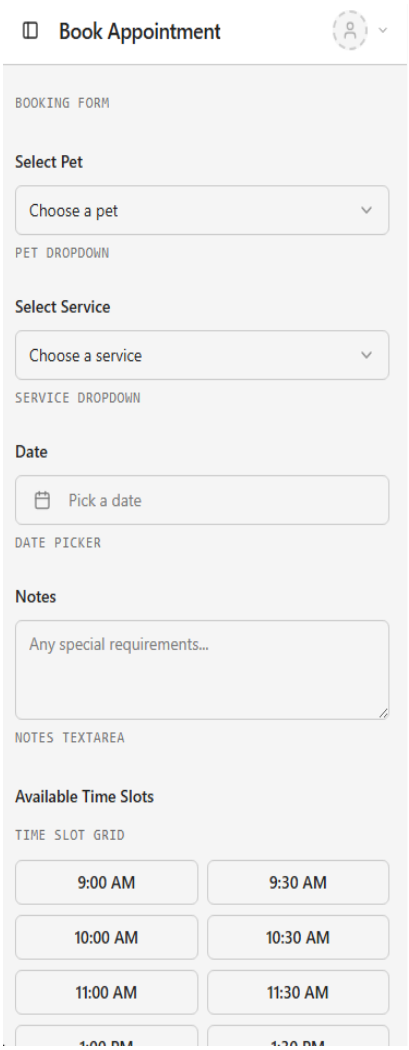
- Clinic Settings:
  - Opening time, closing time, slot length.
  - Save button.
- Blocked Dates:
  - Add date (date picker + Block button).
  - List of blocked dates with Remove buttons.

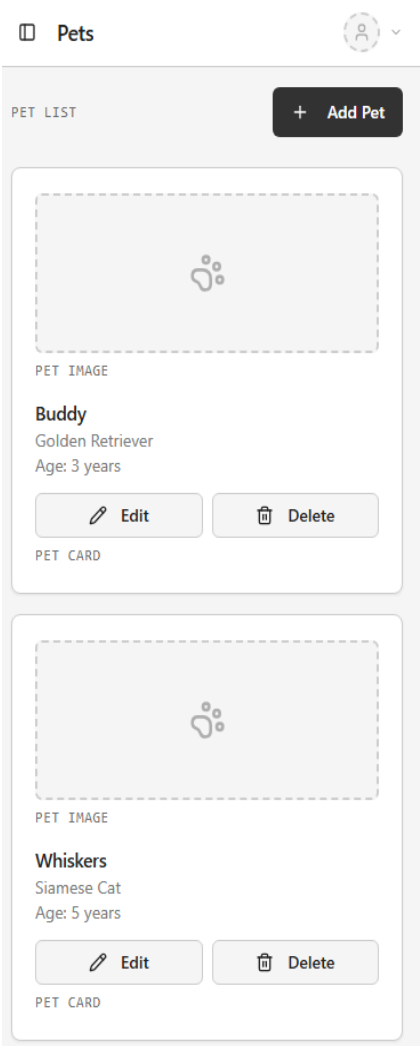
7.2 Mobile Application Wireframes






Dashboard
Next Appointment
APPOINTMENT CARD
Pet: Buddy PET NAME
Service: Annual Checkup SERVICE
Mar 15, 2026 — 10:00 AM DATE/TIME
Confirmed STATUS BADGE
Book Appointment
CTA BUTTON
Profile Summary
PROFILE CARD
John Doe
john@example.com
USER INFO
3
Pets
7
Appointments
STATS


Book Appointment
BOOKING FORM
Select Pet
Choose a pet
PET DROPDOWN
Select Service
Choose a service
SERVICE DROPDOWN
Date
Pick a date
DATE PICKER
Notes
Any special requirements...
NOTES TEXTAREA
Available Time Slots
TIME SLOT GRID
9:00 AM
9:30 AM
10:00 AM
10:30 AM
11:00 AM
11:30 AM


Pets
PET LIST
Add Pet
PET IMAGE
Buddy
Golden Retriever
Age: 3 years
Edit
Delete
PET CARD
PET IMAGE
Whiskers
Siamese Cat
Age: 5 years
Edit
Delete
PET CARD

Pets

Add / Edit Pet

PET FORM MODAL

Name

Pet name

Species

Dog, Cat, etc.

Breed

Select breed

BREED DROPDOWN

Age

Age in years

Notes

Any special notes...

Upload Pet Image

IMAGE UPLOAD

Save

PET CARD

Services

SERVICE CATALOG

Annual Checkup

SERVICE NAME

Complete physical examination and health assessment.

DESCRIPTION

30 min DURATION BADGE

Vaccination

SERVICE NAME

Core and non-core vaccine administration.

DESCRIPTION

15 min DURATION BADGE

Dental Cleaning

SERVICE NAME

Professional dental scaling and polishing.

DESCRIPTION

45 min DURATION BADGE

Surgery Consultation

SERVICE NAME

Pre-surgical evaluation and planning.

My Appointments

APPOINTMENTS LIST

Annual Checkup Confirmed

Date: Mar 15, 2026 Time: 10:00 AM

Pet: Buddy

Reschedule Cancel

APPOINTMENT CARD

Vaccination Pending

Date: Mar 20, 2026 Time: 2:30 PM

Pet: Whiskers

Reschedule Cancel

APPOINTMENT CARD

Dental Cleaning Confirmed

Date: Apr 02, 2026 Time: 11:00 AM

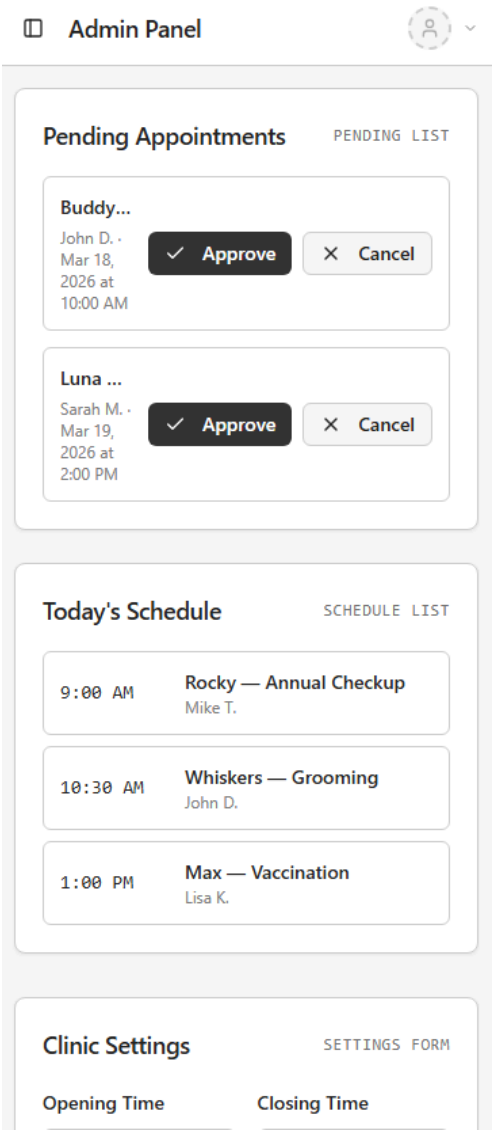
Pet: Rocky

Reschedule Cancel

APPOINTMENT CARD

Grooming Completed

Date: Jan 10, 2026 Time: 9:00 AM


Admin Panel
Pending Appointments
PENDING LIST
Buddy...
John D. · Mar 18, 2026 at 10:00 AM
Approve
Cancel
Luna ...
Sarah M. · Mar 19, 2026 at 2:00 PM
Approve
Cancel
Today's Schedule
SCHEDULE LIST
9:00 AM
Rocky — Annual Checkup
Mike T.
10:30 AM
Whiskers — Grooming
John D.
1:00 PM
Max — Vaccination
Lisa K.
Clinic Settings
SETTINGS FORM
Opening Time
Closing Time

Bottom Navigation

[🏠 Dashboard] [🐾 Pets] [📅 Book] [📋 Appointments] [⚙ Admin] (admin only or nested).

Mobile Dashboard

- Top card: Next appointment.
- Shortcut buttons: Book, View Appointments.

Pets Screen

- List of pets with small cards.
- Floating "+" button to add new pet.

Booking Screen

- Stepper UI:
  - Pet → Service → Date → Time → Notes → Confirm.
- Time slots as chips (scrollable horizontal grid).

Appointments Screen

- List of upcoming appointments:

Admin Screen

- Tabs:
  - Today
  - Pending
  - Settings
- Simple, touch-friendly controls with bigger buttons (min 44x44dp).

Design System

- Colors:
  - Background: #0F0F12
  - Primary: #C67C4E
  - Error: #EF4444
  - Success: #10B981

- Typography: Use Inter/Outfit, responsive sizes.
- Spacing: 8dp grid.
- Components: Reusable cards, chips, badges for statuses.

8.0 PLAN

8.1 Project Timeline (Adapted to VetEase)

Phase 1: Planning & Design (Week 1–2)

- Week 1 – Requirements & Architecture
    - Day 1–2: Project setup, repo structure, initial README.
    - Day 3–4: Functional & non-functional requirements (FRS, SRS).
    - Day 5–7: System architecture diagrams (components, deployment).
- Week 2 – Detailed Design
    - Day 1–2: API specification (endpoints, payloads, error codes).
    - Day 3–4: Database design and ERD.
    - Day 5–6: Web + mobile wireframes.
    - Day 7: Consolidated implementation plan.

Phase 2: Backend Development (Week 3–4)

- Week 3 – Foundation
    - Day 1:
        - Spring Boot project skeleton with dependencies.
        - Database schema creation and ERD implementation
    - Day 2: Database configuration, User entity, security basics.
    - Day 3: JWT authentication, roles, auth endpoints.
    - Day 4: Pet and service entities + repositories.
    - Day 5: Pet and service REST endpoints.

- Week 4 – Scheduling Core
    - Day 1: Appointment entity, repositories, status handling.
    - Day 2: Availability checker logic (slots, blocked dates, no double-booking).
    - Day 3: Appointment booking, cancel, reschedule.
    - Day 4: Admin endpoints (confirm, complete, cancel, today/pending views).
    - Day 5:
        - Google OAuth login integration
        - Pet image upload implementation (file storage service)
        - External API integration (dog breeds lookup)
        - Email notification service implementation (SMTP)

Phase 3: Web Application (Week 5–6)

- Week 5 – Web Foundation
    - Day 1: React + Vite setup; basic routing.
    - Day 2: Auth pages (Login, Register) + integration with backend.
    - Day 3: Dashboard (profile + next appointment).
    - Day 4: Pets management page.
    - Day 5: Services list page.
- Week 6 – Web Features Completion
    - Day 1: Book Appointment page (availability integration).
    - Day 2: My Appointments page (cancel/reschedule).
    - Day 3: Admin Panel (pending + today schedule).
    - Day 4: Admin settings (clinic hours, blocked dates).
    - Day 5:
        - Responsive layout implementation
        - Dark theme polish
        - UI testing and bug fixes

Phase 4: Mobile Application (Week 7–8)

- Week 7 – Android Foundation
  - Day 1: Project setup, navigation skeleton.
  - Day 2: Authentication screens (login/register).
  - Day 3: Dashboard & pets list.
  - Day 4: Booking flow UI.
  - Day 5: Retrofit service layer and DTOs.
- Week 8 – Mobile Feature Completion
  - Day 1: Appointments list + cancel/reschedule.
  - Day 2: Admin-focused screens (if scope allows).
  - Day 3: UI polish and animations.
  - Day 4: Testing on emulator and real devices.
  - Day 5: Final APK build and documentation.

Phase 5: Integration & Deployment (Week 9–10)

- Week 9 – Integration & Testing
  - Day 1: End-to-end flows: register → add pet → book → confirm → complete.
  - Day 2: Cross-platform tests (web & mobile using same backend).
  - Day 3: Security review (roles, JWT expiry, basic hardening).
  - Day 4: Performance sanity tests (availability under multiple bookings).
  - Day 5: Documentation updates (screenshots, user guide).
- Week 10 – Deployment
  - Day 1: Deploy backend (Railway/Heroku-style).
  - Day 2: Deploy web app (Vercel/Netlify).
  - Day 3: Generate and distribute APK.
  - Day 4: Final regression testing.
  - Day 5: Final submission and presentation.

Milestones

- M1 (End Week 2): SRS, SDD, ERD, and wireframes completed.
- M2 (End Week 4): Backend API fully functional with authentication and scheduling.
- M3 (End Week 6): Web application feature-complete.
- M4 (End Week 8): Mobile app feature-complete.
- M5 (End Week 10): System deployed, tested, and documented.

Risk Mitigation

- Implement simplest vertical slice first: login → add pet → book appointment.
- Keep frequent git checkpoints (small, descriptive commits).
- Test backend with tools like Postman early, before UI integration.
- De-prioritize “Could Have” features if schedule slips; ensure “Must Have” is solid.